Equivalent ratios

1 Complete the following pairs of equivalent ratios:

a $1 : 2 = 3 : \square$

b $2 : 5 = \square : 10$

c $3 : 9 = 6 : \square$

d $8 : 4 = \square : 6$

e $2 : \square = 4 : 14$

f $6 : \square = 18 : 12$

g $\square : 20 = 1 : 4$

h $\square : 16 = 8 : 32$

2 Write down three ratios that are equivalent to the following ratios:

a $3 : 1$

b $6 : 10$

c $2 : 5$

d $10 : 100$

3 The following table shows the ratio of dogs to cats:

Dogs	to	Cats
9	:	5
18	:	10
27	:	
45	:	
	:	50

a Complete the table of equivalent ratios.

b If there are 270 dogs, how many cats would there be?

c Simplify the ratio of the number of dogs to cats from part (b).

Simplified ratios

4 Simplify the following ratios:

a $4 : 10$

b $8 : 4$

c $20 : 40$

d $100 : 2000$

e $10 : 20$

f $15 : 40$

g $18 : 22$

h $36 : 48$

i $16 : 24$

j $45 : 35$

k $35 : 5$

l $450 : 350$

5 Simplify the following ratios:

a $\frac{1}{2} : 3$

b $\frac{2}{3} : \frac{1}{2}$

c $\frac{2}{9} : 5$

d $\frac{9}{8} : \frac{5}{8}$

e $\frac{6}{7} : \frac{8}{11}$

f $\frac{3}{4} : \frac{7}{12}$

g $\frac{2}{5} : \frac{3}{4}$

h $\frac{9}{10} : \frac{2}{5}$

i $5\frac{2}{5} : 6$

j $1\frac{5}{7} : 1\frac{9}{11}$

k $1\frac{2}{9} : 3$

l $2\frac{1}{3} : 1\frac{8}{9}$

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6 Simplify the following ratios:

a $7 : 6.3$ **b** $1.5 : 2.4$ **c** $0.5 : 2.25$ **d** $2.4 : 3.9$

e $2.8 : 1.6$ **f** $0.4 : 2.08$ **g** $\$0.60 : \$2.00$ **h** $\$5.20 : \$9.50$

7 Write the following as simplified ratios:

a $\$8.20$ to $\$11.60$ **b** 540c to $\$3.00$

c 5.4 km to 420 m **d** 120 cm to 5.1 m

e 11 years to 28 months **f** 20 days to 5 weeks

8 Write each of the following statements as a simplified ratio:

a There are 11 cats to 6 dogs.

b In a classroom there are 12 girls to 5 boys.

c When making pasta, you need 5 cups of plain flour to 8 eggs.

d There are 3 red marbles to 2 blue marbles.

e The Great Dane weighed two times as much as the Labrador.

f In a working day, a bricklayer lays 50 bricks every hour.

9 Express each pair of quantities as a simplified ratio:

a 9000 metres to 8 kilometres **b** $\$1$ to $31c$

c 2 centuries to 409 years **d** 31 minutes to 2 hours

e 6 minutes to 659 seconds **f** 43 weeks to 1 year

g 37 months to 2 years **h** 13 hours to 5 days

Applications

10 Tricia and Luigi invest money into a business in the ratio $4 : 9$. If Tricia has invested $\$2400$, how much has Luigi invested?

11 The ratio of students to teachers competing in a charity race is $9 : 4$. If 54 students take part in the race, how many teachers are there?

12 Two kinds of pine trees, Bristlecone and Aleppo, are planted in rows. In each row the ratio of Bristlecone to Aleppo is $10 : 9$. Altogether, 600 Bristlecone pine trees are planted. How many Aleppo pines are planted?

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13 At a sporting event, the ratio of security guards to spectators is $1 : 120$. If 8400 spectators attend the event, how many security guards will be required?

14 A painter wants to create a certain colour by mixing two different colours of paint, Vespa and Nitro, in the ratio $4 : 1$. He uses 4 litres of the Nitro colour.

a How many litres of the Vespa colour must he use?

b How many litres of paint will he have altogether once the two colours are combined?

15 Hermione is making a patterned lid for a wooden box. She knows that for every $\frac{3}{5}$ metres of mahogony she will need 70 centimetres of oak.

a She states the ratio of mahogony to oak as $\frac{3}{5} : 70$. Explain why this ratio is incorrect.

b Write the ratio of mahogony to oak using fractions.

c Simplify this ratio.

d How many metres of mahogony would Hermione need if she wants to use 14 metres of oak?

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